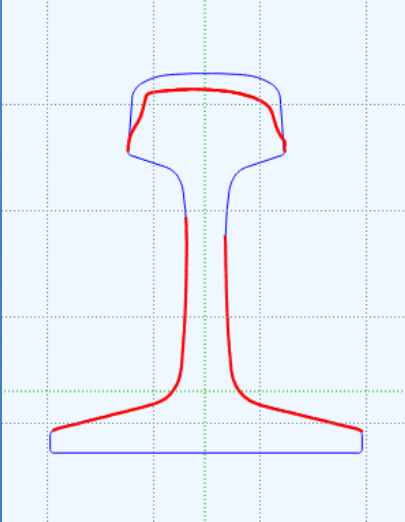




Big Data for Rail Health Assessment: Utilizing Existing Inspection Technologies to Forecast Rail Condition

Michael Palese – Track Engineer
Amtrak

B O S T O N



Introduction



Problem statement: Production rail renewal maintenance planning often relies on qualitative condition data.



Research question: Can rail wear rates be accurately calculated using rail profile cross-sections from TGMS?



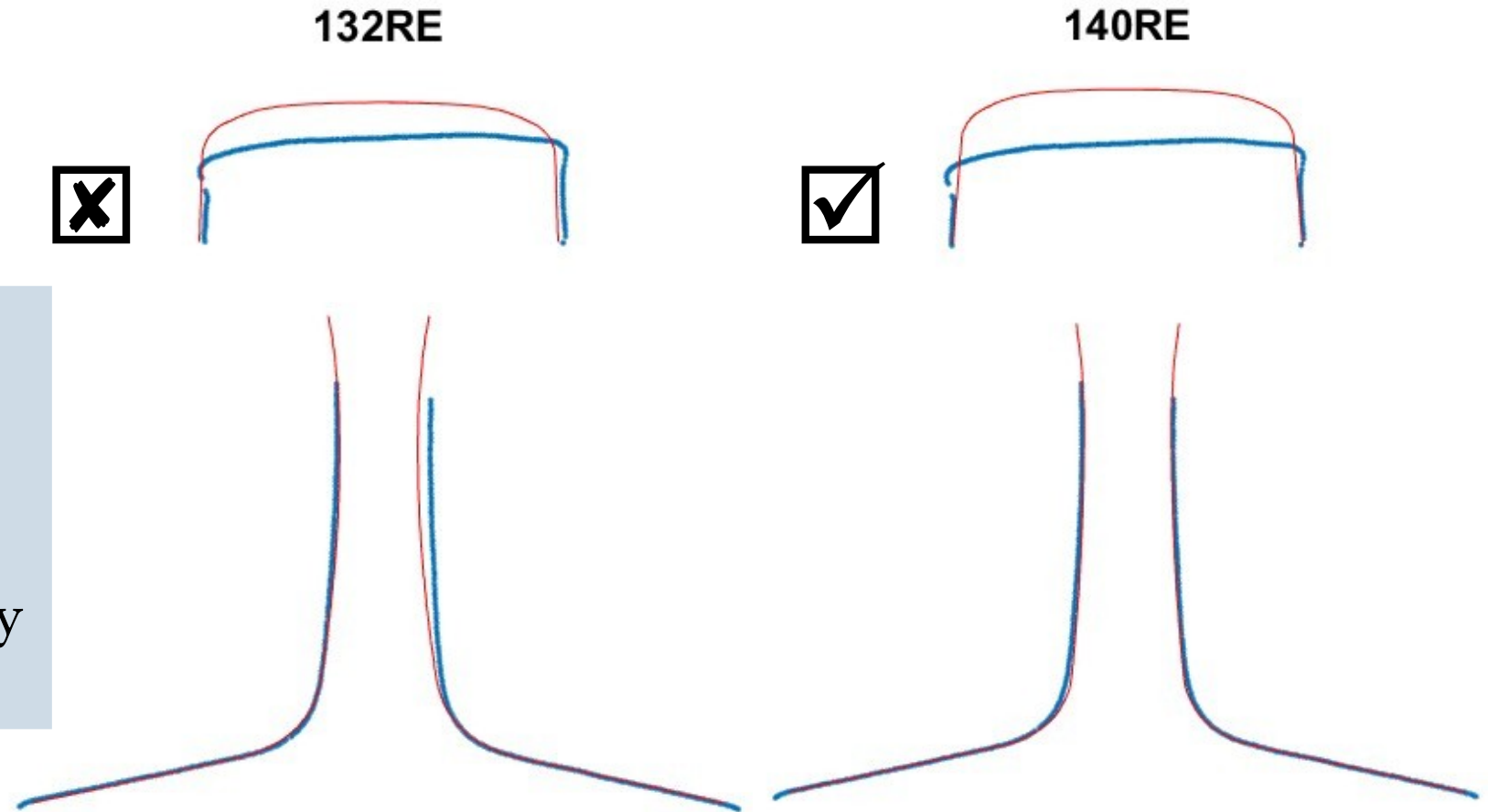
Benefit: Extends the useful-life of rail and allows rail renewal maintenance to be planned further in advance.



Introduction



The rail weight of measured profile cross-sections must be classified before wear metrics can be accurately calculated.





Project workflow



Dataset
assembly

Data pre-
processing

Rail plug
detection

Rail wear
modeling

Rail wear
report



Dataset assembly



Track geometry survey data

- One manned vehicle
- Inspection frequency of track sections varies from once-a-week to twice-a-year.
- Yields:
 - Rail profile scans
 - Geolocation data
 - Curvature data

Geospatial (GIS) data

- Management of asset data and their locations.
- Yields:
 - Track sections (blocks, interlockings, yards, etc.)



Dataset assembly



- Track geometry surveys were reviewed to ensure their quality:
 - Accurately captured rail profile scans.
 - Accurately encoded and georeferenced.
- Resulting dataset:
 - Survey count = 400+
 - Survey date range = April 2019 to Present
- The resulting dataset covers most of Amtrak owned-track sections.





Data pre-processing



Transform

Convert rail profile point clouds into binary images.

Classify

Predict rail weights from profile images using a CNN.

Post-process

Smooth rail weight results to handle poorly classified samples.

Register

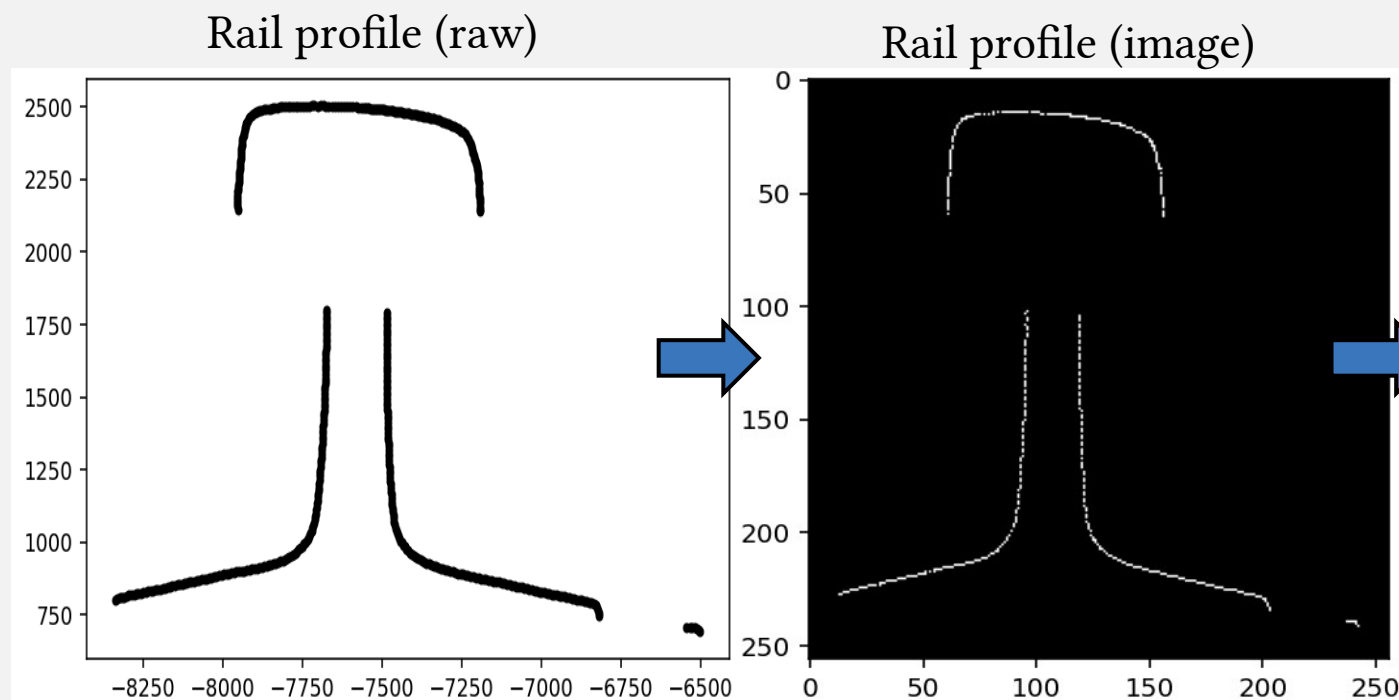
Align rail profiles to matching templates of the same weight.

Wear

Measure wear properties using registered rail profiles.



Transformation



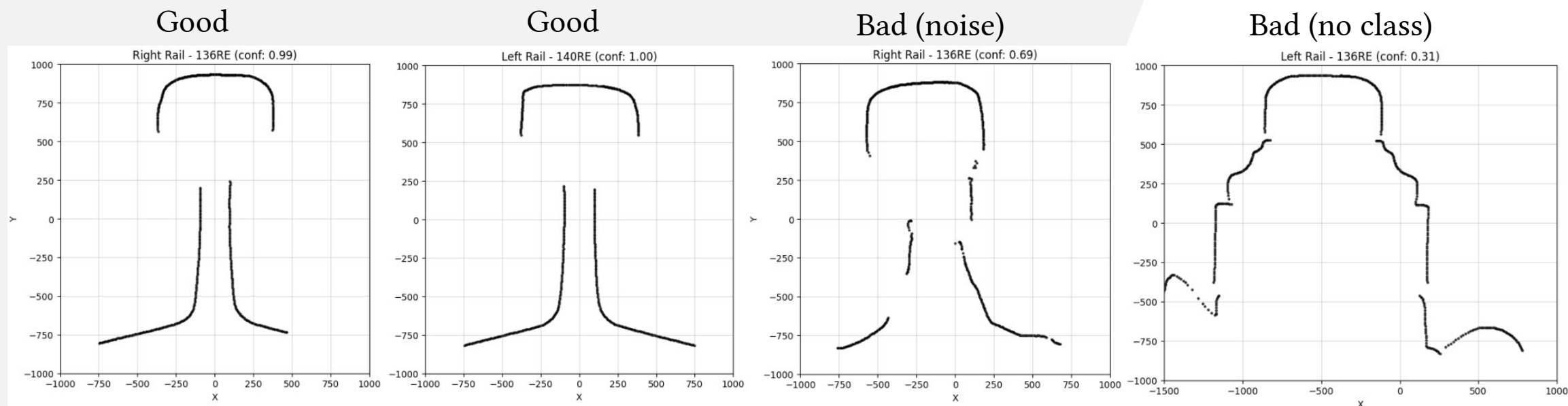
115RE	1.4e-8
119RE	7.6e-6
127DY	3.2e-6
132RE	1.5e-4
136RE	1.5e-3
140RE	0.99
152PS	6.5e-6
155PS	1.3e-6



Classification

Rail profile scans throughout a survey vary in quality:

- Obstructions (snow, dirt, etc.)
- Sunlight glare
- Camera hardware malfunctions



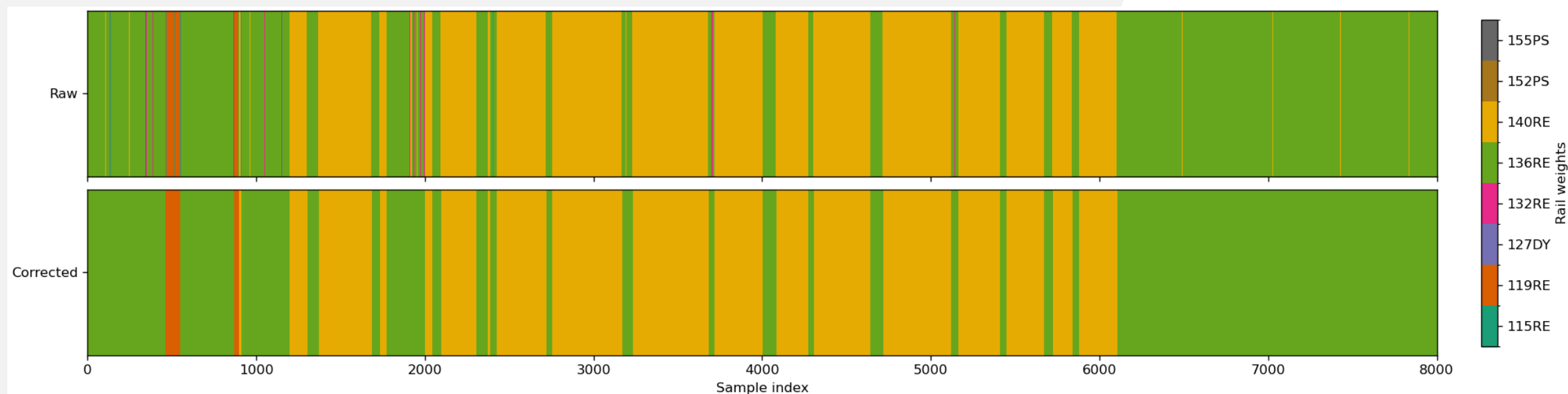


Post-processing



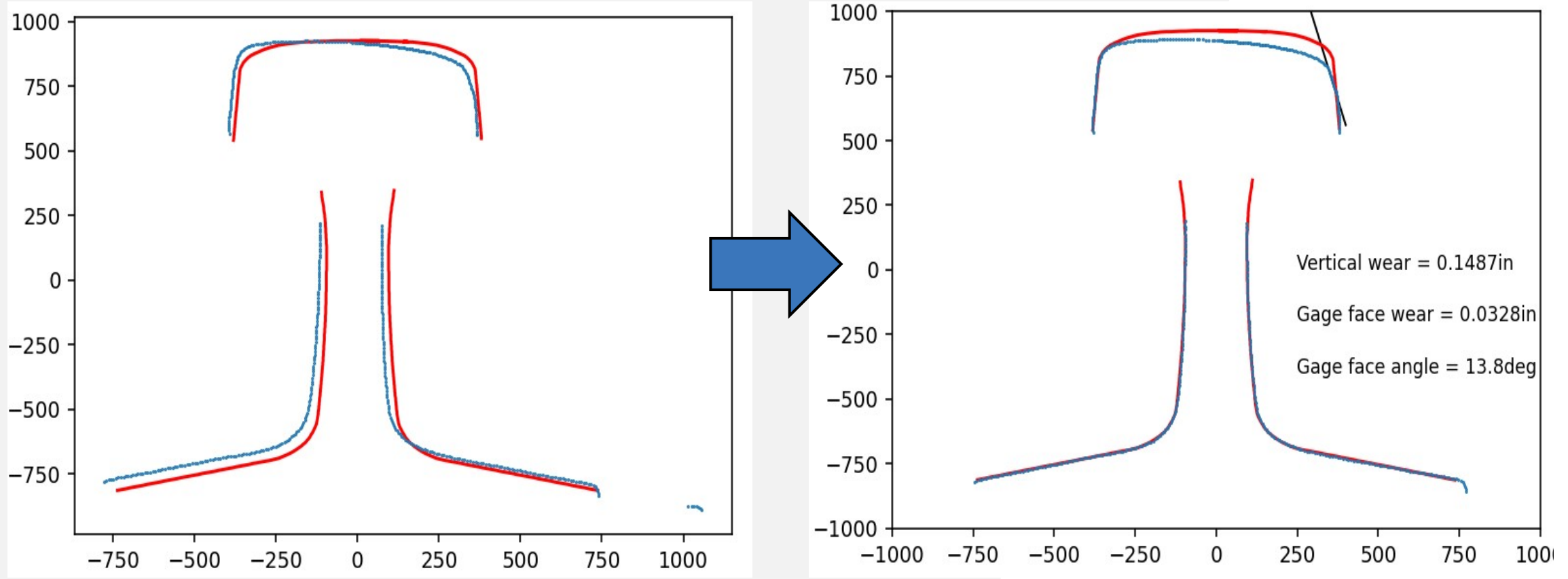
Post-processing smooths classified rail weight sequences:

- Handles poorly classified rail profile scans
- Accounts for special track
- Applies maintenance standards





Registration and wear calculation

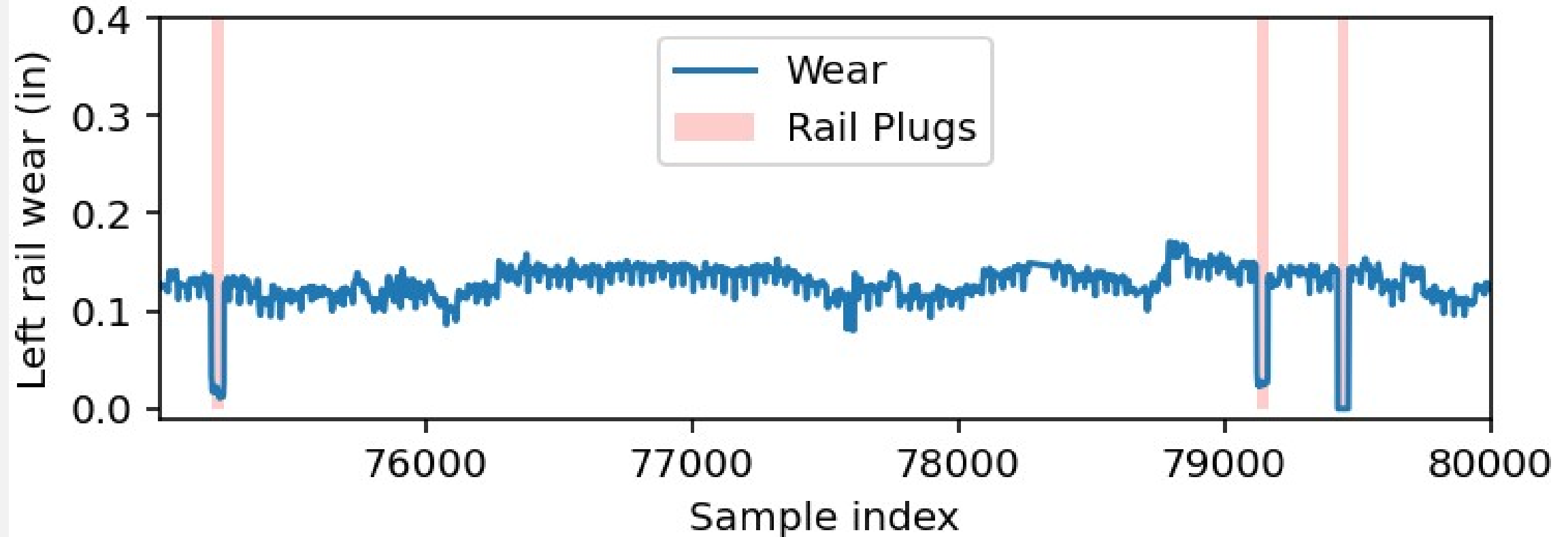




Rail plug detection



Wear signals provide insight about the current health of the rail.





Rail wear model workflow



Identify
surveys
covering
the track
section

Read
survey
data

Create
analytical
segments

Remove
outlier
surveys

Align
surveys

Calculate
wear
statistics

Detect
previous
maintenance
interventions

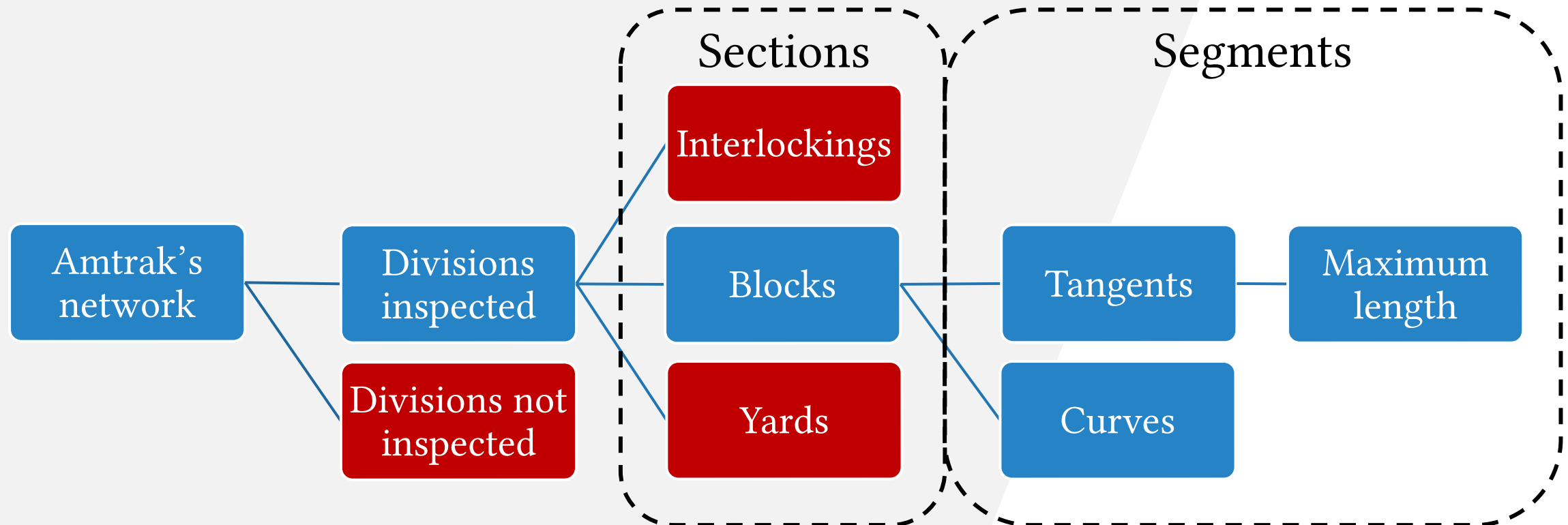
Trend
wear
properties

Merge
segment
features



Data segmentation

The scope of the model is track blocks between interlockings. Each block is further segmented into tangents and curves per maintenance standards.



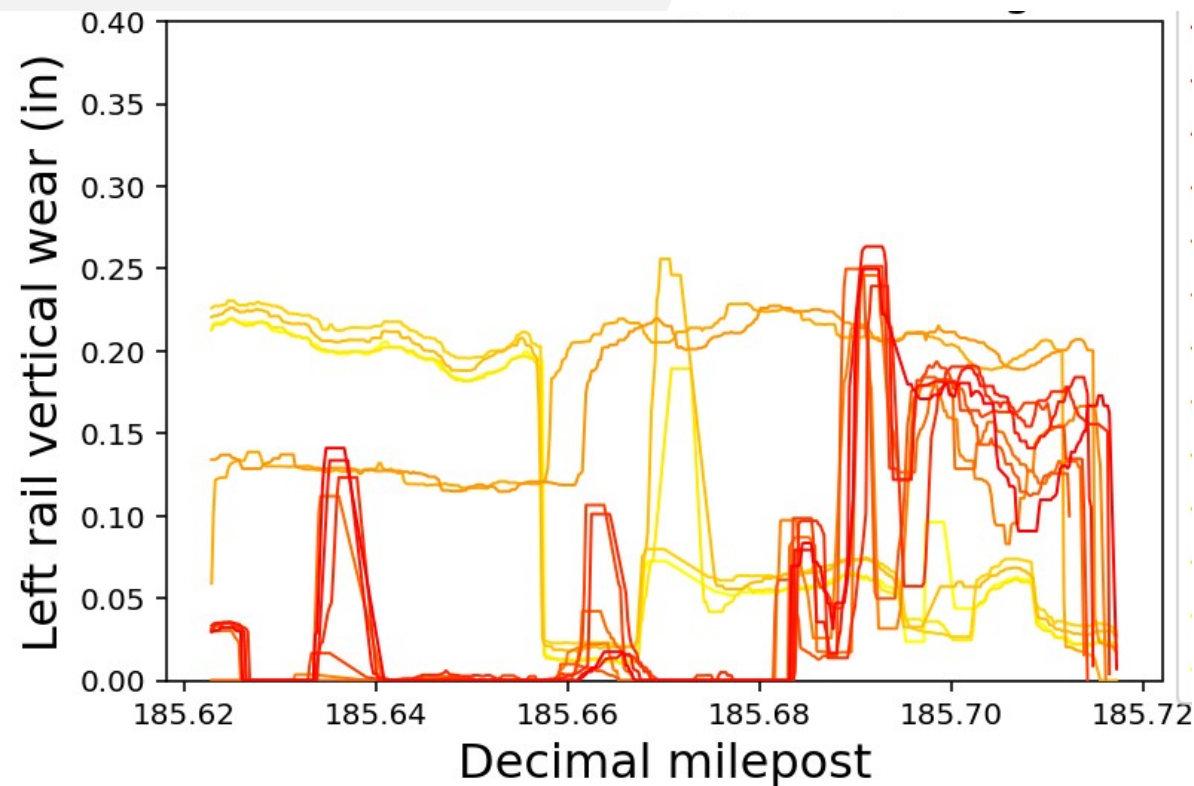
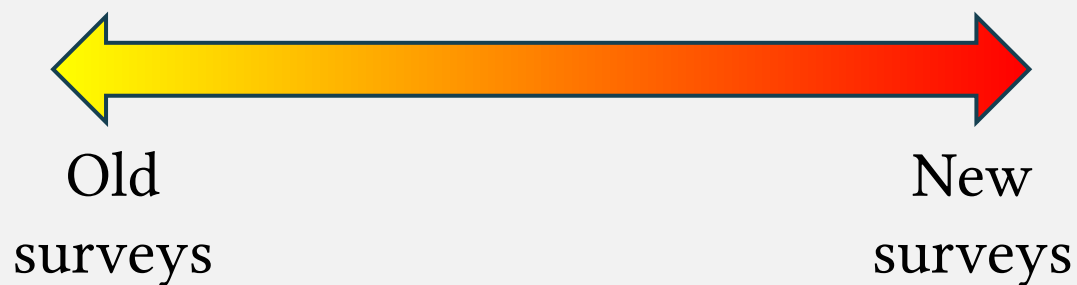


Outlier detection

Finding and removing outliers helps ensure calculated rail wear rates are accurate.

Causes of outlier surveys:

- Poor survey quality
- Issue during inspection
- Incorrectly matched to track section
- Incorrectly matched to track side

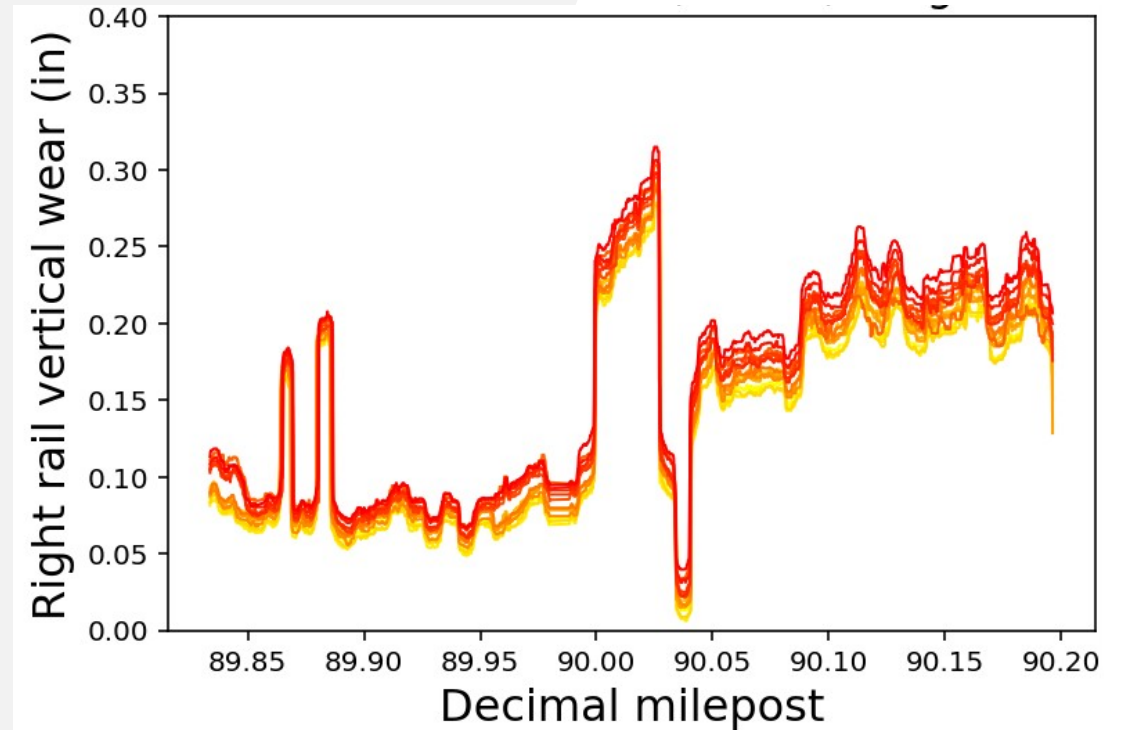
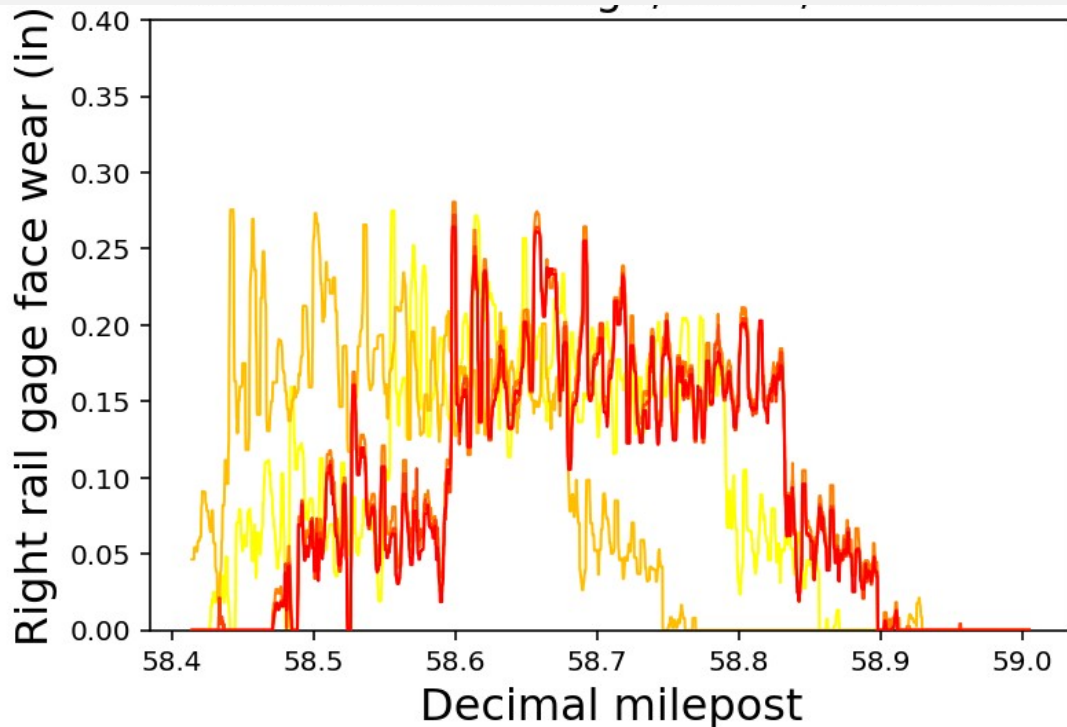




Survey alignment

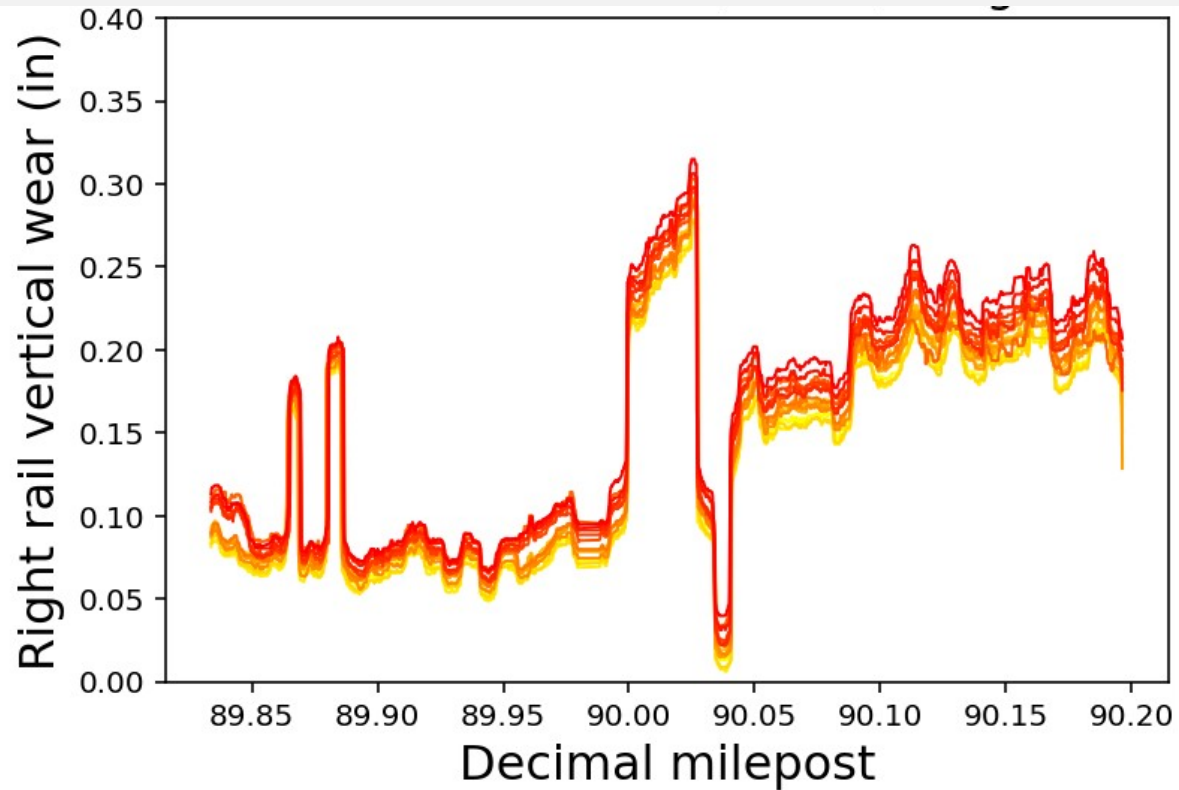


Aligning surveys is done using signal autocorrelation. This ensures accurate wear statistics and foot-by-foot comparisons.





Wear statistics



Wear mean

Wear maximum

Wear standard deviation

Wear gradient

Most common rail weight

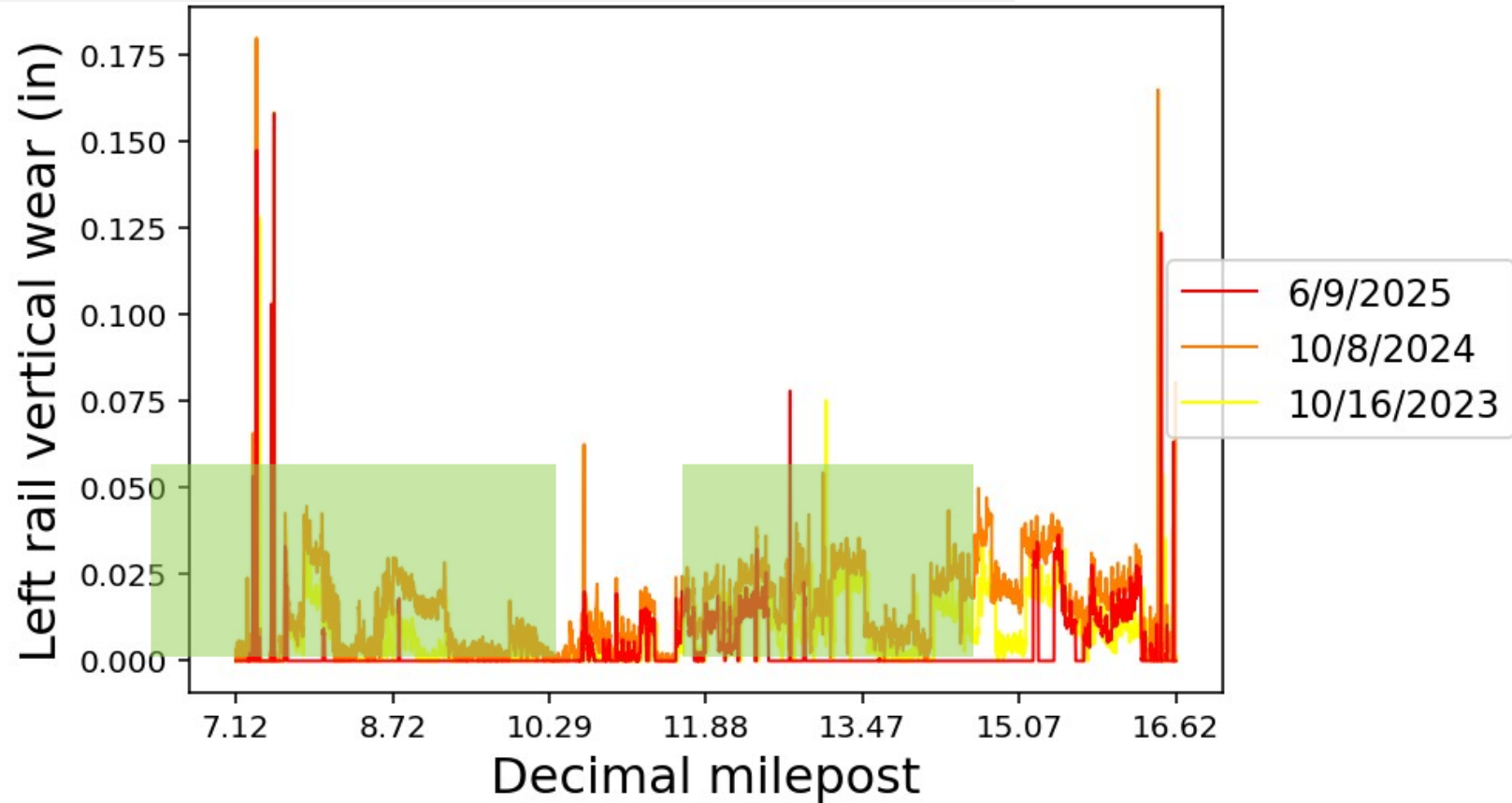
2nd most common rail weight



Maintenance detection

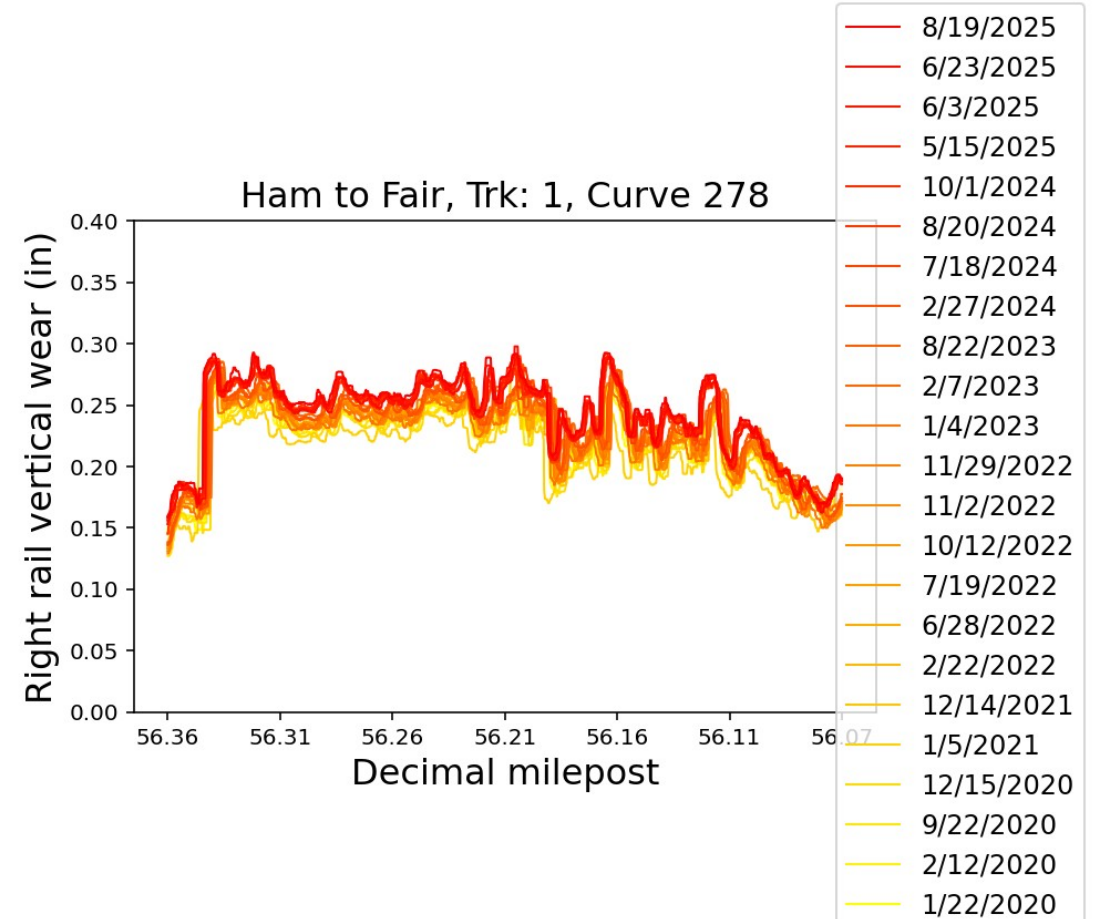
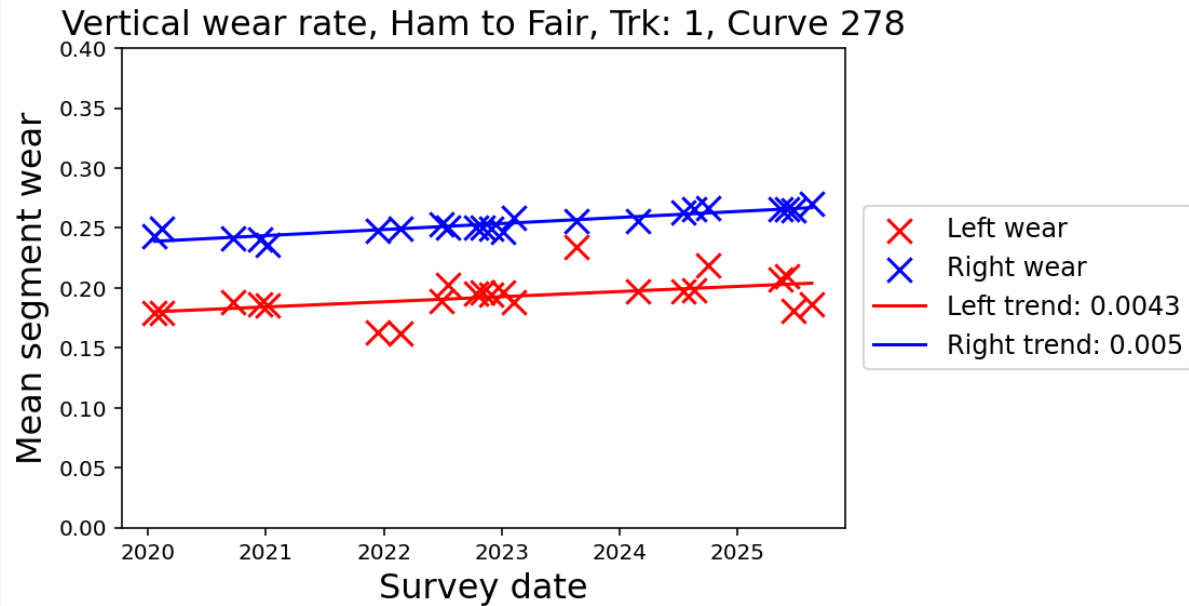


Large variations (drops) in signal statistics approximate maintenance events.



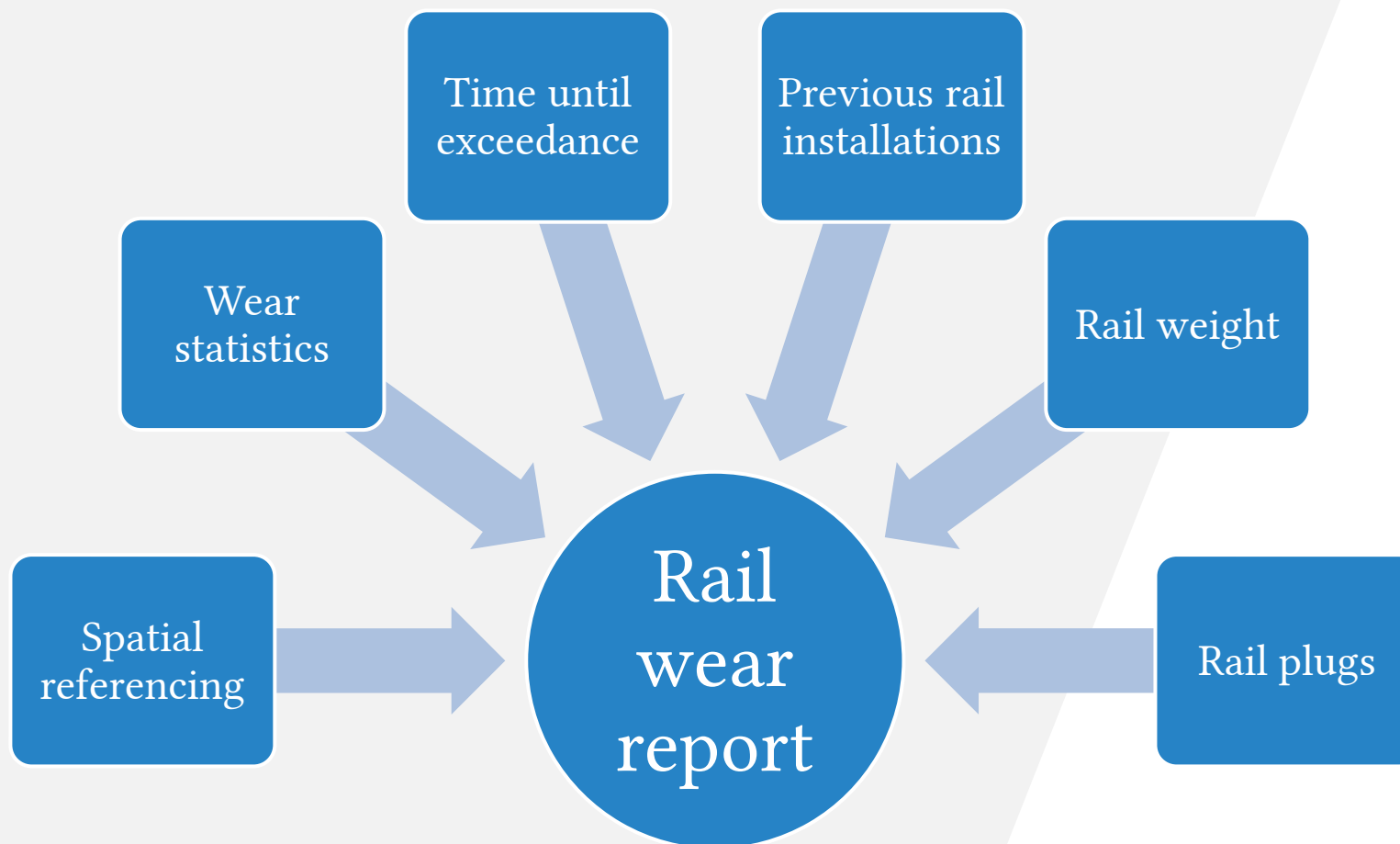


Calculate rail wear rates





Rail wear reporting





Rail wear reporting



SurveyDate	SurveyCount	LineCode	Track #	TrackSide	TrackSide2	MPDir	MPDec_start	MPDec_end	VerticalWear_Current	VerticalWear_Trend	GageFaceWear_Current	GageFaceWear_Trend	Limit	YearsUntilExceedance	RailType
2026-03-30		44 AP	2	Left	East	South	83.92	83.57	0.180	0.005	0.013	-0.003	Compliant	30	140RE
2026-03-17		41 AP	2	Left	East	South	83.56	83.21	0.181	0.004	0.021	-0.001	Compliant	30	140RE
2026-03-30		43 AP	2	Left	East	South	83.21	82.85	0.173	0.004	0.022	-0.002	Compliant	30	140RE
2026-03-30		43 AP	2	Left	East	South	82.85	80.47	0.195	0.005	0.017	0.000	Compliant	20	140RE
2026-03-30		43 AP	2	Left	East	South	80.47	80.13	0.216	0.007	0.054	0.000	Maintenance	30	140RE
2026-03-30		44 AP	2	Left	East	South	80.13	79.79	0.182	0.006	0.051	0.000	Compliant	18	140RE
2026-03-30		23 AP	2	Right	West	South	89.23	88.38	0.000	-0.002	0.002	0.018	Compliant	30	136RE
2026-03-30		23 AP	2	Right	West	South	88.38	88.17	0.000	0.000	0.007	0.013	Compliant	30	136RE
2026-03-30		23 AP	2	Right	West	South	88.17	86.58	0.000	-0.001	0.008	0.018	Compliant	30	136RE
2026-03-30		23 AP	2	Right	West	South	86.58	86.37	0.000	0.000	0.000	0.016	Compliant	30	136RE
2026-03-30		23 AP	2	Right	West	South	86.37	85.72	0.007	-0.001	0.004	0.020	Compliant	30	136RE
2026-03-30		43 AP	2	Right	West	South	85.72	85.36	0.150	0.001	0.023	0.003	Compliant	30	140RE
2026-03-30		43 AP	2	Right	West	South	85.36	85.00	0.144	0.001	0.007	0.002	Compliant	30	140RE
2026-03-30		44 AP	2	Right	West	South	85.00	84.66	0.166	0.001	0.008	0.002	Compliant	30	140RE
2026-03-17		42 AP	2	Right	West	South	84.66	84.30	0.164	0.001	0.030	0.003	Compliant	30	140RE
2026-03-30		44 AP	2	Right	West	South	84.30	83.92	0.160	0.002	0.021	0.003	Compliant	30	140RE
2026-03-30		44 AP	2	Right	West	South	83.92	83.57	0.163	0.000	0.012	0.002	Compliant	30	140RE
2026-03-17		41 AP	2	Right	West	South	83.56	83.21	0.161	0.001	0.006	0.002	Compliant	30	140RE
2026-03-30		43 AP	2	Right	West	South	83.21	82.85	0.161	0.001	0.008	0.001	Compliant	30	140RE
2026-03-30		43 AP	2	Right	West	South	82.85	80.47	0.170	0.001	0.059	0.001	Compliant	30	140RE
2026-03-30		43 AP	2	Right	West	South	80.47	80.13	0.184	0.002	0.029	0.002	Compliant	30	140RE
2026-03-30		44 AP	2	Right	West	South	80.13	79.79	0.163	0.003	0.023	0.003	Compliant	30	140RE
2026-03-30		43 AP	2	Left	East	South	89.92	89.75	0.328	0.005	0.082	-0.005	Safety	0	140RE
2026-03-30		43 AP	2	Left	East	South	89.75	89.72	0.305	0.006	0.009	0.000	Safety	0	140RE
2026-03-30		44 AP	2	Left	East	South	89.72	89.56	0.303	0.006	0.016	0.000	Safety	0	140RE
2026-03-30		43 AP	2	Right	West	South	89.92	89.75	0.281	0.001	0.000	0.000	Maintenance	19	140RE
2026-03-30		43 AP	2	Right	West	South	89.75	89.72	0.281	0.001	0.000	0.001	Maintenance	13	140RE
2026-03-30		44 AP	2	Right	West	South	89.72	89.56	0.288	0.003	0.027	0.000	Maintenance	4	140RE
2026-03-30		39 AP	2	Left	East	South	126.47	126.31	0.199	0.002	0.046	0.000	Compliant	30	140RE
2026-03-30		37 AP	2	Left	East	South	126.31	125.94	0.225	0.006	0.168	-0.003	Maintenance	30	140RE
2026-03-30		37 AP	2	Left	East	South	125.94	125.57	0.173	0.003	0.049	0.000	Compliant	30	140RE
2026-03-30		37 AP	2	Left	East	South	125.57	125.20	0.208	0.006	0.219	-0.001	Maintenance	30	140RE
2026-03-04		38 AP	2	Left	East	South	125.20	124.85	0.158	0.006	0.038	0.001	Compliant	22	140RE
2026-02-12		38 AP	2	Left	East	South	124.84	124.50	0.162	0.006	0.025	-0.001	Compliant	30	140RE



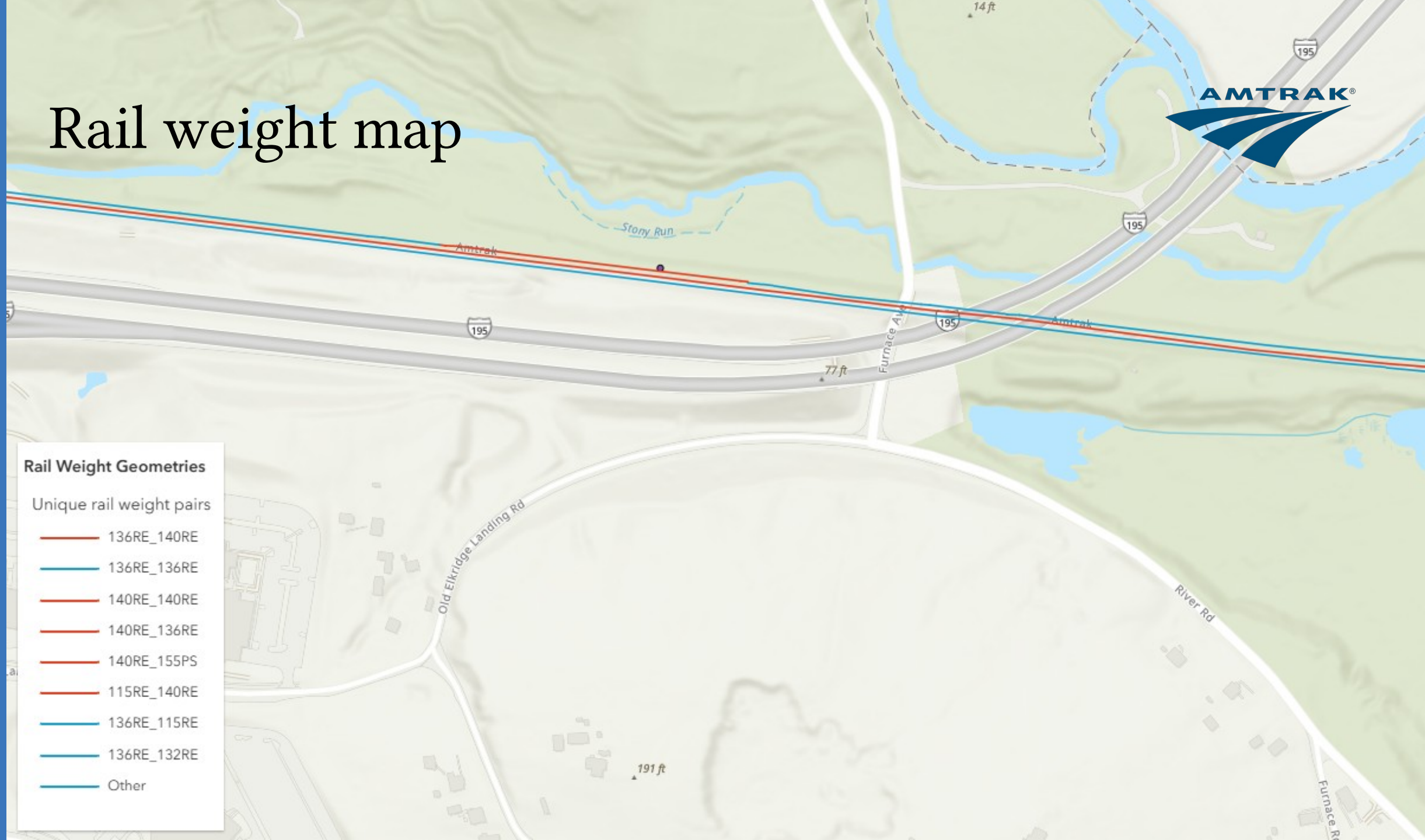
Rail weight map



Rail Weight Geometries

Unique rail weight pairs

- 136RE_140RE
- 136RE_136RE
- 140RE_140RE
- 140RE_136RE
- 140RE_155PS
- 115RE_140RE
- 136RE_115RE
- 136RE_132RE
- Other





Conclusion



- Track surveys and track asset data sufficiently represent rail wear condition.
- Rail profile scans can be used to classify the weight rail.
- Frequent track surveys allow maintenance to be detected, including plug installations, and rail wear condition to be forecasted.
- Ongoing work:
 - Improving CNN to predict presence of special trackwork.
 - Using rail profile head contour to support rail grinding need.
 - Improving segmentation to model wear in interlockings, stations and yards.
 - Correlating internal rail flaw defects analytical segments in the rail wear report to better represent all aspects of rail health.
 - Developing better visualization tools to support maintenance scheduling.

Thanks!

